

Correlation and Regression 2

Review

$$\begin{matrix} 1 & 2 & 3 & 4 & 5 \\ \frac{1}{3} & \frac{4}{15} & \frac{1}{5} & \frac{2}{15} & \frac{1}{5} \end{matrix}$$

$$\begin{aligned} E(X) &= \frac{7}{3} \\ \text{Var}(X) &= \frac{229}{15} - \frac{49}{9} \\ &= \frac{14}{9} \end{aligned}$$

$$\begin{aligned} E(2X) &= \frac{14}{3} \\ \text{Var}(2X) &= 4 \times \frac{14}{9} = \frac{56}{9} \end{aligned}$$

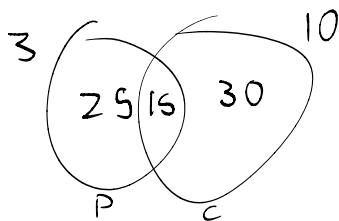
$$\begin{aligned} E(3X+4) &= 11 \\ \text{Var}(3X+4) &= 14 \end{aligned}$$

Z	A	B	C	D	E	F	G	H	I	J	K	L
5	9	8	6	10	2	9	3	1	7	12	11	4
5	2	6	3	12	4	5	8	7	1	9	10	11

$$3 \ 2 \ 3 \ 2 \ 2 \ 4 \ 5 \ 6 \ 6 \ 3 \ 1 \ .7$$

$$11 \ 4 \ 11 \ 4 \ 4 \ 16 \ 25 \ 36 \ 36 \ 9 \ 1 \ 41$$

$$1 - \frac{6 \times 202}{12 \times 143} = 0.2937$$



$$\frac{25}{30} = \frac{5}{6}$$

Consolidation

$$\bar{x} = \frac{1+2+3+4+5+6+7+8}{8} \quad \sum x = 36$$

$$= \frac{1}{2} = 4.5$$

$$\begin{aligned} \bar{y} &= \frac{1.5+6+7.7+7.3+9.1+8.6+10.6+10.2}{8} \\ &= \frac{64}{8} \quad \sum y = 64 \\ &= 8 \end{aligned}$$

$$\begin{aligned} \text{ii) } \sum xy &= 321.7 & S_{xy} &= 321.7 - \frac{64 \times 36}{8} = 33.7 \\ \sum x^2 &= 204 & S_{xx} &= 204 - \frac{36^2}{8} = 42 \end{aligned}$$

$$\begin{aligned} \text{Regression line: } y &= bx+a = \frac{S_{xy}}{S_{xx}}x + (\bar{y} - b\bar{x}) \\ &= \frac{33.7}{42}x + 8 - \frac{33.7}{42} \times 4.5 \\ &= 0.802x + 4.39 \end{aligned}$$

$$a) y = 0.802 \times 5 + 4.39 = 7.197$$

$$b) y = 0.802 \times 11 + 4.39 = 13.212$$

$$iii) y = 0.802x + 4.39$$

$$x = \frac{y - 4.39}{0.802}$$

$$= \frac{9 - 4.39}{0.802} = 2.56$$

$$2) y = a + bx$$

$$S_{xy} = 64123 - \frac{660 \times 1270}{10} = 303$$

$$S_{xx} = 43678 - \frac{660^2}{10} = 118$$

$$= \bar{y} - \frac{S_{xy}}{S_{xx}} \bar{x} + \frac{S_{xy}}{S_{xx}} x$$

$$\bar{y} = 127$$

$$\bar{x} = 66$$

$$y = 127 - 2.57 \times 66 + 2.57x$$

$$= 2.57x - 42.47$$

$$@x = 64, y = 122.16$$

$$3) \sum x = 800 \quad \sum y = 1380 \quad \sum xy = 64092$$

$$\sum x^2 = 26888 \quad \sum y^2 = 184088$$

$$S_{xx} = 26888 - \frac{26000}{10} = 1888$$

$$S_{yy} = 184088 - \left(\frac{1380^2}{10} = 182250 \right) = 1838$$

$$S_{xy} = 64092 - \left(\frac{800 \times 1380}{10} = 67200 \right) = 1592$$

$$r = \frac{1592}{\sqrt{1888 \times 1838}}$$

$$= 0.855$$

$$ii) \bar{x} = 50$$

$$\bar{y} = 138$$

$$\text{regression line: } y = \left(\bar{y} - \frac{S_{xy}}{S_{xx}} \bar{x} \right) + \frac{S_{xy}}{S_{xx}} x$$

$$= 138 - \frac{1592}{1888} \times 50 + \frac{1592}{1888} x$$

$$= 92.8 + 0.843x$$

$$iii) y = 92.8 + 0.843 \times 63$$

$$= 124$$

Exam Questions

1 a) Independent

$$\begin{aligned} \sum v &= 270 & \sum d &= 291 & n &= 6 \\ \sum v^2 &= 13900 & \sum vd &= 18970 \end{aligned}$$

$$S_{vv} = 13900 - \frac{270^2}{6} = 1750$$

$$S_{vd} = 18970 - \frac{270 \times 291}{6} = 2875$$

$$b = \frac{S_{vd}}{S_{vv}} = \frac{2875}{1750} = 1.643$$

$$\begin{aligned} a &= \bar{d} - b\bar{v} \\ &= \frac{291}{6} - 1.643 \times \frac{270}{6} \end{aligned}$$

$$= -25.435$$

$$d = 1.643v - 25.435$$

@ $v = 45$

$$d = 48.5$$

c) reliable as price close to 1 so less error when extrapolation
lower bikes may get more with speed,
people may be more wary @ higher speeds
so faster reaction time.

2 a) one is likely to depend on the other because there is a general positive correlation between c and t observable on the graph

b) $r = \frac{S_{ct}}{S_c S_t}$

$$\begin{aligned} \sum t &= 232 & \sum c &= 260 & \sum xy &= 7489 \\ \sum t^2 &= 6130 & \sum c^2 &= 9444 & n &= 9 \\ \bar{t} &= 25.78 & \bar{c} &= 31.11 \end{aligned}$$

$$S_c = 9444 - \frac{260^2}{9} = 732.89$$

$$S_{tt} = 6130 - \frac{232^2}{9} = 149.56$$

$$S_{ct} = 7489 - \frac{232 \times 260}{9} = 211.22$$

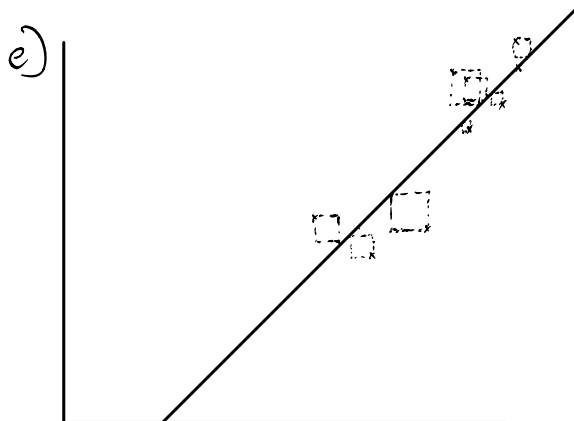
$$\therefore r = 0.819$$

c) r would be the same: 0.819.

This is because the variable ' t ' would have undergone a linear transformation ($1.6t + 32$) so r would not change.

d) regression line: $c = a + bt$

$$\begin{aligned} &= (\bar{c} - \frac{S_{ct}}{S_{tt}} \bar{t}) + \frac{S_{ct}}{S_{tt}} t \\ &= 31.11 - 1.813 \times 25.78 + 1.813t \\ &= 1.813t - 15.64 \end{aligned}$$



The line drawn is the line which minimizes the area of the squares created from the points.